

PATIENT A · WEEKS 1 TO 8

Camila Reyes, 19

19 years old. Second year kinesiology student and a midfielder on the college soccer team. No prior medical history, no medications.

You have her for the first half of the term. You will know her better than you expect to.

STUDENT: _____ ENTRY POINT: _____

Face sheet

Who the patient is, and the two things about you that stay the same all term.

PATIENT	YOUR NAME
Camila Reyes, 19	

HISTORY
19 years old.
Second year
kinesiology
student and
a midfielder
on the
college
soccer
team. No
prior medical
history, no
medications.

ALLERGIES
None known.

MEDICATIONS
BEFORE THIS
CHART
STARTS

None at
baseline.

YOUR
ENTRY
POINT,
PICKED
IN
WEEK 1
AND
KEPT

☐ Nursing

☐ Medicine

☐ Radiology

☐ Respiratory
therapy

Exercise
☐ and allied
health

—

**BASELINE,
AUGUST 12,
PRESEASON
PHYSICAL**

Heart rate

52 · **Blood**

pressure

108/64 ·

Mean

arterial

pressure 79 ·

Respiratory

rate 12 ·

Temperature

36.8 (98.2) ·

Oxygen

saturation

99% RA ·

Weight kg

(lb) 61 (134) ·

Sodium 139 ·

Potassium

4.2 ·

Chloride 103

.

Bicarbonate	
25	Glucose
88	
Creatinine	
0.8	
Hemoglobin	
13.4	
Hematocrit	
40	

—

One column per week, in course order. The date under each week is when that entry happened in the story; the weeks are not in date order, and that is on purpose. The baseline column is filled in for you.

[illegible]

STUDENT: _____ ENTRY POINT: _____

Intake and output flowsheet

Record what the chart gives you, and use the last two rows to record what it does not. Weight is on the vitals sheet; when the two disagree, say which you believe.

Measure	Wk 1 August 12 Preseason physical	Wk 2 September 18 Athletic training room, then the emergency department	Wk 3 September 22, 0640 Arrival in the emergency department	Wk 4 September 22, 0700 to 1100 Resuscitation, hours 0 to 4	Wk 5 September 22, and October 20 Neurological, reflex and sensory assessment	Wk 6 September 25 to October 10 Mobility assessment and follow-up	Wk 7 September 22 to December 5 Endocrine, autonomic and reproductive follow-up	Wk 8 December 5 Discharge summary and case closure
Oral or enteral in								
Intravenous in								
TOTAL IN								
Urine out								
Other measured out								
TOTAL OUT								
Net for the period								
Running total since admission								
Losses not on this chart								

Write the units the first time a test appears. Blank rows are for a test a week introduces that is not listed.

[illegible]

A problem goes on the list the day you can name it. Write the week you found it and the week it resolved, if it does. Number them; the numbers are how you refer to them on the week pages.

[illegible]

Fluids, drugs, oxygen, procedures. What was given, when in the story, what it was meant to change, and whether it did.

[illegible]

STUDENT: _____ ENTRY POINT: _____

01 Preseason physical

August 12 · Six weeks before she got sick · Foundations of Physiology

Nothing is wrong yet. This is the only chart entry in the term where every number is normal, which is exactly what makes it the most useful one. Everything you read for the next fourteen weeks gets compared to this page.

YOUR ENTRIES

NEW ON THE CHART THIS WEEK

Copy the numbers into the flowsheets first, then list here what changed since the last column.

PROBLEM LIST, ADDITIONS AND CHANGES

Refer to problems by the numbers you gave them on the problem list.

DRAW THE CONTROL LOOP

Regulated variable, sensor, integrator, effector, and where this week it is failing or being pushed.

THE FIVE QUESTIONS

1. Name three variables in this chart that are actively regulated. For each one, say what quantity is being held steady and what control system is holding it.
2. Her resting heart rate is 52. Say whether that is a finding or a baseline, and say exactly what you would need to know to decide.
3. Explain what a reference range is a range of. Then say why a value inside it is not proof of health, and a value outside it is not proof of disease.
4. Interpret this chart. Say what it shows and what it cannot show.

5. Name the one measurement you would add to this baseline today, and say what question it would let you answer in October.

FROM YOUR DESK

Your entry point's data and question are on this week's entry page. Answer it here.

STUDENT: _____ ENTRY POINT: _____

02 Athletic training room, then the emergency department

September 18 · Five weeks after her preseason physical · The Cell, Transport and Water

She did the right thing as she understood it. Two sessions in the heat, and she drank water at every break, about five liters across the day, because she had been told not to get dehydrated. She finished the second session confused and vomiting. Her weight is up, not down. Everything in this entry turns on one idea: water follows solute, and it is the solute that cannot cross that decides where the water goes.

YOUR ENTRIES

NEW ON THE CHART THIS WEEK

Copy the numbers into the flowsheets first, then list here what changed since the last column.

PROBLEM LIST, ADDITIONS AND CHANGES

Refer to problems by the numbers you gave them on the problem list.

DRAW THE CONTROL LOOP

Regulated variable, sensor, integrator, effector, and where this week it is failing or being pushed.

THE FIVE QUESTIONS

1. Draw her three fluid compartments before practice and after, with volumes and with osmolarity marked on each. Show the water movement between them with arrows, and say which compartment her 2.5 kg went into and which one made her confused.

2. She drank only water and her sodium fell to 124. Explain the mechanism in terms of what she lost in sweat, what she replaced it with, and why replacing water alone lowers a concentration rather than restoring it.

3. Look at the three microscopy slides. Name the tonicity of each solution relative to the cell, say which way water moved and why, and name the one property of the solute that decides the answer in every case.

4. She is given 3% saline, not normal saline and not free water. Predict what each of those three fluids would do to her cell volume, and say why the one chosen is the one that helps.

5. Her urine osmolality is 380 mOsm/kg while her serum is 256. Say what a healthy kidney would be doing with a serum osmolality that low, say what hers is doing instead, and name the one measurement that would tell you whether her kidney is the problem or is being told to do this.

FROM YOUR DESK

Your entry point's data and question are on this week's entry page. Answer it here.

STUDENT: _____ ENTRY POINT: _____

03 Arrival in the emergency department

September 22, 0640 · Four days after the training room · Catch Up on the Cell

Nothing new opens this week, and this entry opens nothing new either. It is the same five ideas you worked with last week, which solute cannot cross, which way water goes, what that does to cell volume, and what a signal does when its receptor is empty, put in front of a much harder case. Four days ago she was confused on a field with a sodium of 124. This morning a teammate brought her in. Everything you need is either on this page or was on last week's.

RESULTS RELEASED THIS WEEK

BASIC METABOLIC PANEL

Collected September 22, 0645

Test	Result	Units	Reference range	Flag
Sodium (measured)	128	mEq/L	135 to 145	Low
Potassium	5.4	mEq/L	3.5 to 5.0	High
Chloride	96	mEq/L	98 to 107	Low
Bicarbonate	4	mEq/L	22 to 29	Critical low
Glucose	642	mg/dL	70 to 99	Critical high
Blood urea nitrogen	34	mg/dL	7 to 20	High
Creatinine	1.6	mg/dL	0.5 to 1.0	High

OSMOLALITY

Collected September 22, 0645

Test	Result	Units	Reference range	Flag
Serum osmolality (measured)	305	mOsm/kg	275 to 295	High
Urine osmolality	320	mOsm/kg	no fixed range, read against serum	

YOUR ENTRIES

NEW ON THE CHART THIS WEEK

Copy the numbers into the flowsheets first, then list here what changed since the last column.

PROBLEM LIST, ADDITIONS AND CHANGES

Refer to problems by the numbers you gave them on the problem list.

DRAW THE CONTROL LOOP

Regulated variable, sensor, integrator, effector, and where this week it is failing or being pushed.

THE FIVE QUESTIONS

1. Name the compartments involved and say which solute is setting the gradient between them. Say what makes that solute able to set a gradient at all.
2. Predict which way water has moved between the intracellular and extracellular compartments, and say what has happened to cell volume. Then say why she looks dry on the outside while a specific compartment inside her is being pulled dry too.
3. Her glucose is 642 mg/dL and her cells have no fuel. Name the transporter that cannot reach the cell membrane without insulin, name two tissues that take up glucose without needing it, and explain how a bloodstream full of glucose and a starving muscle cell are the same fact.
4. Correct her sodium for the glucose. Say which correction factor you used and that more than one is in use. Then interpret the measured 128: say what it shows and what it cannot show.
5. Name the one measurement that would change your fluid plan in the next hour, and say what you would do differently for a high result and for a low one.

FROM YOUR DESK

Your entry point's data and question are on this week's entry page. Answer it here.

STUDENT: _____ ENTRY POINT: _____

04 Resuscitation, hours 0 to 4

September 22, 0700 to 1100 · The first four hours of treatment · Membrane Potential, Neurons and Synapses

This week is about potassium and her nervous system. When she arrives, the potassium in her blood is high. Four hours later it is low. Her body did not take in potassium and then lose it in those four hours. Most of the change is potassium moving between her blood and the inside of her cells, so the blood number on its own can fool you. At the same time, her nervous system is working hard. Her heart is racing, the blood vessels in her skin are squeezed shut, and her skin is dry. All three come from the same part of her nervous system, the sympathetic division, but it does not use the same chemical messenger and receptor for each job.

RESULTS RELEASED THIS WEEK

POTASSIUM SERIES

Collected September 22

Test	Result	Units	Reference range	Flag
Potassium, 0700	5.4	mEq/L	3.5 to 5.0	High
Potassium, 0900	4.1	mEq/L	3.5 to 5.0	
Potassium, 1100	3.1	mEq/L	3.5 to 5.0	Low
Magnesium, 0700	1.6	mg/dL	1.7 to 2.4	Low
Phosphate, 0700	4.8	mg/dL	2.5 to 4.5	High

YOUR ENTRIES

NEW ON THE CHART THIS WEEK

Copy the numbers into the flowsheets first, then list here what changed since the last column.

PROBLEM LIST, ADDITIONS AND CHANGES

Refer to problems by the numbers you gave them on the problem list.

DRAW THE CONTROL LOOP

Regulated variable, sensor, integrator, effector, and where this week it is failing or being pushed.

THE FIVE QUESTIONS

1. Which ion is this entry about? Across a resting cell membrane, which way does it tend to move, and why? What keeps its concentration difference in place, and what does that cost the cell?

2. Predict what a blood potassium of 5.4 does to the resting membrane potential. Then predict what 3.1 does. For each, say which way the resting potential moves, and say which one puts her in more danger at 1100.

3. A cell that stays partly depolarized becomes harder to fire, not easier. Explain why, one gate at a time, and name the channel state that causes it.

4. Her heart is fast and the blood vessels in her skin are squeezed shut. For each one, name the neurotransmitter and the receptor involved, and explain how the same part of the nervous system can cause two different effects. Then look at her dry skin. What does it tell you about her sympathetic nervous system, and what can it not tell you?

5. What one lab result or event would make you stop the insulin for now? Why is stopping it the right move, even though her blood sugar is still high?

FROM YOUR DESK

Your entry point's data and question are on [this week's entry page](#). Answer it here.

STUDENT: _____ ENTRY POINT: _____

05 Neurological, reflex and sensory assessment

September 22, and October 20 · The first morning, and four weeks later · Reflexes, and Sensing the World

Three loops are on this page and none of them needed a doctor to run. One protects her brain, one defends her blood pressure, and one changed the shape of a lens without anyone noticing until she stopped wearing her contact lenses. Two of the three recover on their own. The interesting question is what you would have to measure to know which.

RESULTS RELEASED THIS WEEK

BEDSIDE SENSORY AND REFLEX TESTING

Collected September 22, 0810

Test	Result	Units	Reference range	Flag
Patellar reflex, both sides	1+		2+ is the usual normal	Low
Ankle reflex, both sides	1+		2+ is the usual normal	Low
Plantar response	flexor		flexor	
Vibration, 128 Hz at both great toes	felt		felt	
10 g monofilament, 9 sites per foot	felt at 9 of 9		felt at 9 of 9	
Distance acuity, both eyes	20/60		20/20	Low
Pupil size and reaction	4 mm, brisk and equal		2 to 5 mm, brisk	

YOUR ENTRIES

NEW ON THE CHART THIS WEEK

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PROBLEM LIST, ADDITIONS AND CHANGES

Refer to problems by the numbers you gave them on the problem list.

DRAW THE CONTROL LOOP

Regulated variable, sensor, integrator, effector, and where this week it is failing or being pushed.

THE FIVE QUESTIONS

1. Name the five parts of a reflex arc. Then map her pupillary light reflex onto them, naming the actual structure at each step, and say which part of that arc is the one you are testing when you shine a light in one eye and watch the other.

2. Her blood pressure fell from 96/58 to 78/44 when she sat up, and her heart rate rose by 10. Name the reflex that should have prevented the fall, name its sensor and where that sensor sits, and say why the response you can see on this page is not enough.

3. Her distance vision was 20/60 on September 22 and 20/20 four weeks later, with no change in prescription and nothing done to her eyes. Explain the mechanism using what you know about water moving toward solute, name the structure that changed shape, and say why the change took weeks to reverse when her glucose came down in hours.

4. Interpret reflexes of 1+ at 0640 and 2+ at 1400. Say what that change shows and what it cannot show.

5. Name the one sensory test you would add today that would change what you tell her about the next twenty years, and say what a normal result would and would not rule out.

FROM YOUR DESK

Your entry point's data and question are on [this week's entry page](#). Answer it here.

STUDENT: _____ ENTRY POINT: _____

06 Mobility assessment and follow-up

September 25 to October 10 · Hospital day 4 through two weeks after discharge · Muscle, and How Movement Gets Commanded

She is out of danger and she cannot get out of a chair without pushing off. Three weeks of catabolism and four days of lying still cost her more than she expects, and the recovery does not run in the order she expects either. Her force comes back before her muscle does, which tells you the command and the machinery are two separate things.

RESULTS RELEASED THIS WEEK

MUSCLE CHEMISTRY

Collected September 25

Test	Result	Units	Reference range	Flag
Creatine kinase	340	U/L	30 to 190	High
Urine myoglobin	not detected		not detected	
Potassium	4.0	mEq/L	3.5 to 5.0	
Calcium (corrected)	9.1	mg/dL	8.5 to 10.2	
Phosphate	3.2	mg/dL	2.5 to 4.5	

YOUR ENTRIES

NEW ON THE CHART THIS WEEK
Copy the numbers into the flowsheets first, then list here what changed since the last column.

PROBLEM LIST, ADDITIONS AND CHANGES
Refer to problems by the numbers you gave them on the problem list.

DRAW THE CONTROL LOOP
Regulated variable, sensor, integrator, effector, and where this week it is failing or being pushed.

THE FIVE QUESTIONS

1. Name every link in the chain from the motor cortex to a crossbridge. Then say where in that chain her weakness sits, and say what evidence in this chart rules each of the other links in or out.

2. Predict what has happened to her maximal force and to her endurance, separately, and say whether they had to change by the same amount.

3. Explain excitation contraction coupling from the arrival of the action potential at the muscle fiber to the crossbridge, naming the structures and the ion movements. Then say which step is limited when there are simply fewer crossbridges available.

4. Interpret a grip strength of 22 kg against a baseline of 32 kg. Say what it shows and what it cannot show.

5. Name the one measurement that would separate loss of muscle from loss of the ability to activate the muscle she still has, and say what each result would mean.

FROM YOUR DESK

Your entry point's data and question are on this week's entry page. Answer it here.

STUDENT: _____ ENTRY POINT: _____

07 Endocrine, autonomic and reproductive follow-up

September 22 to December 5 · Admission through the eleven week clinic visit · Hormones, the Autonomic System and Reproduction

This is the entry where her diagnosis is actually made, and where the hormones that made her so sick get named. It is also where a thyroid result moves twice without her thyroid ever being the problem, and where a period that stopped in August turns out to be a message from her hypothalamus about how much she has been eating.

RESULTS RELEASED THIS WEEK

REPRODUCTIVE AXIS

Collected October 20

Test	Result	Units	Reference range	Flag
Follicle stimulating hormone	2.4	IU/L	3.5 to 12.5 (follicular)	Low
Luteinizing hormone	1.8	IU/L	2.4 to 12.6 (follicular)	Low
Estradiol	18	pg/mL	30 to 100 (follicular)	Low
Prolactin	11	ng/mL	4 to 23	
TSH	2.1	mIU/L	0.4 to 4.0	

PELVIC ULTRASOUND

Collected October 20

Test	Result	Units	Reference range	Flag
Endometrial thickness	3	mm	8 to 12 before ovulation	Low
Largest follicle	7	mm	18 to 24 when dominant	Low
Ovarian volume, right	6.2	mL	3 to 10	
Ovarian volume, left	5.8	mL	3 to 10	

YOUR ENTRIES

NEW ON THE CHART THIS WEEK

Copy the numbers into the flowsheets first, then list here what changed since the last column.

PROBLEM LIST, ADDITIONS AND CHANGES

Refer to problems by the numbers you gave them on the problem list.

DRAW THE CONTROL LOOP

Regulated variable, sensor, integrator, effector, and where this week it is failing or being pushed.

THE FIVE QUESTIONS

1. Name each hormone in this entry, the tissue it acts on, and whether its receptor sits on the cell surface or inside the cell. Say what that location predicts about how quickly each one can change anything.
2. Adrenaline appears in this chart twice: as a hormone and as the end of a nerve. Draw the sympathetic pathway to her heart and the sympathetic pathway to her adrenal medulla side by side, naming every transmitter and receptor. Say what is structurally different about the second one, and why that difference makes one effect fast and local and the other slower and everywhere.
3. Predict which way glucagon, cortisol, growth hormone and adrenaline each moved on arrival, and say why they all moved in the same direction at once. Then name what was missing from the loop that would normally have limited them.
4. Interpret her October reproductive panel: FSH 2.4, LH 1.8, estradiol low, ovaries normal on ultrasound. Say what it shows and what it cannot show, and contrast it point by point with a panel showing high FSH and low estradiol.
5. Name the one measurement that would change her insulin plan in the next month, and say what a high and a low result would each make you do.

FROM YOUR DESK

Your entry point's data and question are on this week's entry page. Answer it here.

STUDENT: _____ ENTRY POINT: _____

08 Discharge summary and case closure

December 5 · Eleven weeks after discharge, reading the whole file · Closing Camila's File

Nothing new opens this week and nothing new is asked. This is the week of Midterm 1, and this entry is the review: seven chart entries about one person, read as one person. Then you hand her over, because in Week 9 you pick up a patient who is nothing like her.

RESULTS RELEASED THIS WEEK

DECEMBER 5 REVIEW PANEL

Collected December 5

Test	Result	Units	Reference range	Flag
Sodium	140	mEq/L	135 to 145	
Potassium	4.3	mEq/L	3.5 to 5.0	
Chloride	104	mEq/L	98 to 107	
Bicarbonate	25	mEq/L	22 to 29	
Glucose (fasting)	118	mg/dL	70 to 99	High
Creatinine	0.8	mg/dL	0.5 to 1.0	
Hemoglobin A1c	7.1	%	under 5.7 (non-diabetic)	High
Hemoglobin	13.1	g/dL	12.0 to 15.5	
Hematocrit	39	%	36 to 46	

YOUR ENTRIES

NEW ON THE CHART THIS WEEK

Copy the numbers into the flowsheets first, then list here what changed since the last column.

PROBLEM LIST, ADDITIONS AND CHANGES

Refer to problems by the numbers you gave them on the problem list.

DRAW THE CONTROL LOOP

Regulated variable, sensor, integrator, effector, and where this week it is failing or being pushed.

THE FIVE QUESTIONS

1. Pick any three entries from Weeks 1 to 7 and name one regulated variable in each. For each one, name the sensor, the integrator and the effector, and say whether the loop failed, was overwhelmed, or worked correctly and was misread by the people looking at it.
2. Her potassium was 5.4 while her total body potassium was depleted by roughly 200 mEq. Her measured sodium was 128 while her corrected sodium was 137. Her serum was full of glucose while her muscle cells had no fuel. Name the single idea all three share, and say what it means for how you read any number on any chart.
3. Trace one molecule of glucose from her September 22 bloodstream to the inside of a muscle cell, and say exactly where the journey stops. Then trace the same molecule on December 5 and say what is standing in for the missing step.
4. Interpret this file as a whole. Say what seven entries about one patient showed you that seven entries about seven patients could not have, and say what a file like this one cannot show.
5. Name the one measurement you would have added to her August preseason physical, and say what question it would have let you answer in October.

FROM YOUR DESK

Your entry point's data and question are on this week's entry page. Answer it here.

PATIENT B · WEEKS 9 TO 15

Dale Whitcomb, 68

68 years old. Retired long haul driver. Anterior heart attack in 2020 with a stent, heart failure since, ejection fraction 30 percent in March. High blood pressure. Type 2 diabetes for 14 years. Chronic kidney disease, baseline creatinine 1.5 mg/dL. Smoked for 40 years, stopped in 2019.

Camila had one thing wrong at a time. He has four, and they are all talking to each other. That is why he arrives in Week 9 and not before.

STUDENT: _____ ENTRY POINT: _____

Face sheet

Who the patient is, and the two things about you that stay the same all term.

PATIENT

Dale Whitcomb, 68

YOUR NAME

HISTORY

68 years
old.

Retired
long haul
driver.

Anterior
heart
attack in
2020
with a
stent,
heart
failure
since,
ejection

fraction
30
percent
in March.
High
blood
pressure.
Type 2
diabetes
for 14
years.
Chronic
kidney
disease,
baseline
creatinine
1.5 mg/
dL.
Smoked
for 40
years,
stopped
in 2019.

ALLERGIES

None known.

**MEDICATIONS
BEFORE THIS
CHART
STARTS**

Carvedilol 12.5
mg twice daily.
Lisinopril 10 mg
daily.
Furosemide 40
mg daily.
Spironolactone
25 mg daily.
Atorvastatin 40
mg daily.
Metformin
1,000 mg
twice daily.
Insulin glargine
22 units at
night. Aspirin 81
mg daily.

YOUR
ENTRY
POINT,
PICKED
IN
WEEK 1
AND
KEPT

☐ Nursing

☐ Medicine

☐ Radiology

☐ Respiratory
therapy

Exercise
☐ and allied
health

—

**BASELINE,
MARCH
10,
CLINIC
VISIT,
WELL**

Heart rate
64 · **Blood**
pressure
118/70 ·
Mean
arterial
pressure 86 ·
Respiratory

rate 16 ·

Temperature

36.8 (98.2) ·

Oxygen

saturation

96% RA ·

Weight kg

(lb) 88 (194) ·

Sodium 138 ·

Potassium

4.4 ·

Chloride 104

·

Bicarbonate

26 · **Glucose**

142 ·

Creatinine

1.5 · **Blood**

urea

nitrogen 28 ·

Albumin 3.8 ·

Hemoglobin

12.6 ·
Hematocrit
38 · **White**
cells 7.2 ·
Platelets
220 ·
Lactate 1.1

—

STUDENT: _____ ENTRY POINT: _____

Vitals flowsheet

One column per week, in course order. The date under each week is when that entry happened in the story; the weeks are not in date order, and that is on purpose. The baseline column is filled in for you.

Measure	Baseline March 10 Clinic visit, well	Wk 9 March 10, and November 4 at 0840 Emergency department, arrival	Wk 10 November 4, 0900 to 1800 Resuscitation and hemodynamic support	Wk 11 November 4 to November 8 The infection, and the response to it	Wk 12 November 8 to November 16 Nutrition, metabolism and the gut	Wk 13 November 5 to November 10 Respiratory failure, ventilation and weaning	Wk 14 November 4 to November 18 Acute kidney injury on chronic disease	Wk 15 Both files, read again Integration
Heart rate	64							
Blood pressure	118/70							
Mean arterial pressure	86							
Respiratory rate	16							
Temperature °C (°F)	36.8 (98.2)							
Oxygen saturation	96% RA							
Oxygen delivered								
Weight kg (lb)	88 (194)							
Conscious level								

STUDENT: _____ ENTRY POINT: _____

Intake and output flowsheet

Record what the chart gives you, and use the last two rows to record what it does not. Weight is on the vitals sheet; when the two disagree, say which you believe.

Measure	Wk 9 March 10, and November 4 at 0840 Emergency department, arrival	Wk 10 November 4, 0900 to 1800 Resuscitation and hemodynamic support	Wk 11 November 4 to November 8 The infection, and the response to it	Wk 12 November 8 to November 16 Nutrition, metabolism and the gut	Wk 13 November 5 to November 10 Respiratory failure, ventilation and weaning	Wk 14 November 4 to November 18 Acute kidney injury on chronic disease	Wk 15 Both files, read again Integration
Oral or enteral in							
Intravenous in							
TOTAL IN							
Urine out							
Other measured out							
TOTAL OUT							
Net for the period							
Running total since admission							
Losses not on this chart							

STUDENT: _____ ENTRY POINT: _____

Lab flowsheet

Write the units the first time a test appears. Blank rows are for a test a week introduces that is not listed.

Measure	Baseline March 10 Clinic visit, well	Wk 9 March 10, and November 4 at 0840 Emergency department, arrival	Wk 10 November 4, 0900 to 1800 Resuscitation and hemodynamic support	Wk 11 November 4 to November 8 The infection, and the response to it	Wk 12 November 8 to November 16 Nutrition, metabolism and the gut	Wk 13 November 5 to November 10 Respiratory failure, ventilation and weaning	Wk 14 November 4 to November 18 Acute kidney injury on chronic disease	Wk 15 Both files, read again Integration
Sodium mEq/L	138							
Potassium mEq/L	4.4							
Chloride mEq/L	104							
Bicarbonate mEq/L	26							
Anion gap								
Anion gap corrected for albumin								
Glucose mg/dL	142							
Blood urea nitrogen mg/dL	28							
Creatinine mg/dL	1.5							
Albumin g/dL	3.8							
Lactate mmol/L	1.1							
pH								
PaCO ₂ mmHg								
PaO ₂ mmHg								
PaO ₂ to inspired oxygen ratio								
Hemoglobin g/dL	12.6							
Hematocrit %	38							
White cells ×10 ⁹ /L	7.2							
Platelets ×10 ⁹ /L	220							
Phosphate mg/dL								
Magnesium mg/dL								

Measure	Baseline March 10 Clinic visit, well	Wk 9 March 10, and November 4 at 0840 Emergency department, arrival	Wk 10 November 4, 0900 to 1800 Resuscitation and hemodynamic support	Wk 11 November 4 to November 8 The infection, and the response to it	Wk 12 November 8 to November 16 Nutrition, metabolism and the gut	Wk 13 November 5 to November 10 Respiratory failure, ventilation and weaning	Wk 14 November 4 to November 18 Acute kidney injury on chronic disease	Wk 15 Both files, read again Integration

STUDENT: _____ ENTRY POINT: _____

Hemodynamic and monitoring flowsheet

Numbers that come from a monitor or an echocardiogram rather than from a tube of blood.

Measure	Wk 9 March 10, and November 4 at 0840 Emergency department, arrival	Wk 10 November 4, 0900 to 1800 Resuscitation and hemodynamic support	Wk 11 November 4 to November 8 The infection, and the response to it	Wk 12 November 8 to November 16 Nutrition, metabolism and the gut	Wk 13 November 5 to November 10 Respiratory failure, ventilation and weaning	Wk 14 November 4 to November 18 Acute kidney injury on chronic disease	Wk 15 Both files, read again Integration
Cardiac output L/min							
Stroke volume mL							
Central venous pressure mmHg							
Systemic vascular resistance							
Central venous O ₂ saturation %							
Ejection fraction %							
Urine output mL/h							

A problem goes on the list the day you can name it. Write the week you found it and the week it resolved, if it does. Number them; the numbers are how you refer to them on the week pages.

[illegible]

Fluids, drugs, oxygen, procedures. What was given, when in the story, what it was meant to change, and whether it did.

[illegible]

STUDENT: _____ ENTRY POINT: _____

09 Emergency department, arrival

March 10, and November 4 at 0840 · His last well clinic visit, and the morning he arrived · The Heart as a Pump

He has had a bad pump for six years and he has lived around it. Then he got an infection, which asks a heart for more output, and his heart has none left to give. The trap on this page is that his cardiac output reads normal. Read it against what a man with a temperature of 38.9 °C (102.0 °F) should be producing and it is not normal at all, and two numbers on this page tell you so.

RESULTS RELEASED THIS WEEK

CHEMISTRY

Collected November 4, 0850

Test	Result	Units	Reference range	Flag
Sodium	132	mEq/L	135 to 145	Low
Potassium	5.1	mEq/L	3.5 to 5.0	High
Chloride	98	mEq/L	98 to 107	
Bicarbonate	15	mEq/L	22 to 29	Low
Glucose	268	mg/dL	70 to 99	High
Blood urea nitrogen	48	mg/dL	7 to 20	High
Creatinine	2.6	mg/dL	0.7 to 1.3	High
Albumin	2.9	g/dL	3.5 to 5.0	Low

CARDIAC AND PERFUSION MARKERS

Collected November 4, 0850

Test	Result	Units	Reference range	Flag
Troponin T (high sensitivity)	68	ng/L	under 14	High
B-type natriuretic peptide	1,840	pg/mL	under 100 (his March value was 480)	High
Lactate	4.6	mmol/L	0.5 to 2.0	High
Central venous oxygen saturation	52	%	70 to 80	Low

ARTERIAL BLOOD GAS, ON 6 L NASAL CANNULA

Collected November 4, 0855

Test	Result	Units	Reference range	Flag
pH	7.32		7.35 to 7.45	Low
PaCO ₂	30	mmHg	35 to 45	Low
PaO ₂	62	mmHg	80 to 100	Low
Bicarbonate	15	mEq/L	22 to 26	Low
SaO ₂	92	%	95 to 100	Low

YOUR ENTRIES

NEW ON THE CHART THIS WEEK

Copy the numbers into the flowsheets first, then list here what changed since the last column.

PROBLEM LIST, ADDITIONS AND CHANGES

Refer to problems by the numbers you gave them on the problem list.

DRAW THE CONTROL LOOP

Regulated variable, sensor, integrator, effector, and where this week it is failing or being pushed.

THE FIVE QUESTIONS

1. Cardiac output has four inputs. Name them, and for each one say which number on this page reports it and which way it has moved since March.
2. His cardiac output is 4.9 L/min, which sits inside the normal adult range. Say what it should be in a 68 year old man with a temperature of 38.9 °C (102.0 °F) and an infection, show where your estimate comes from, and use the difference to say whether his pump is keeping up. Then name the two numbers on this page that are the evidence it is not.
3. Explain the Frank-Starling relationship. Draw his curve against a normal one, mark where he was in March and where he is this morning, and use the drawing to predict what another liter of fluid would do to his stroke volume.
4. Interpret an ejection fraction of 25 percent. Say what it shows and what it cannot show. Then work out his end diastolic volume from the numbers on this page, and use it to explain why his stroke volume is only a third below normal when his ejection fraction is less than half of normal.
5. Name the one measurement that would change what you give him in the next hour, and say what you would do differently for a high result and for a low one.

FROM YOUR DESK

Your entry point's data and question are on this week's entry page. Answer it here.

STUDENT: _____ ENTRY POINT: _____

10 Resuscitation and hemodynamic support

November 4, 0900 to 1800 · Day 1, the first nine hours · Pressure, Flow, and Holding Blood Pressure Steady

Mean arterial pressure is the product of flow and resistance, and a body can defend it by raising either one. Over nine hours he is asked to do both, and the chart records exactly what each one costs him. The fluid that was meant to help him made his lung worse within twenty-five minutes, and the drug that raised his pressure did not raise his flow.

RESULTS RELEASED THIS WEEK

DERIVED HEMODYNAMICS

Collected November 4

Test	Result	Units	Reference range	Flag
Mean arterial pressure, 0900	58	mmHg	70 to 100	Low
Mean arterial pressure, 1100	73	mmHg	70 to 100	
Systemic vascular resistance, 0900	718	dyn·s·cm ⁻⁵	800 to 1,200	Low
Systemic vascular resistance, 1100	814	dyn·s·cm ⁻⁵	800 to 1,200	
Systemic vascular resistance, 1800	671	dyn·s·cm ⁻⁵	800 to 1,200	Low
Cardiac index, 1800	3.2	L/min/m ²	2.5 to 4.0	

SERUM ALBUMIN AND ONCOTIC PRESSURE

Collected November 4

Test	Result	Units	Reference range	Flag
Albumin	2.9	g/dL	3.5 to 5.0	Low
Estimated plasma colloid osmotic pressure	13	mmHg	25 to 28	Low

YOUR ENTRIES

NEW ON THE CHART THIS WEEK

Copy the numbers into the flowsheets first, then list here what changed since the last column.

PROBLEM LIST, ADDITIONS AND CHANGES

Refer to problems by the numbers you gave them on the problem list.

DRAW THE CONTROL LOOP

Regulated variable, sensor, integrator, effector, and where this week it is failing or being pushed.

THE FIVE QUESTIONS

1. Write the equation connecting mean arterial pressure, cardiac output and systemic vascular resistance. Calculate his systemic vascular resistance at 0900 and at 1100 from the numbers on this page, show the arithmetic, and say in one sentence what the noradrenaline actually changed.

2. He was given 500 mL. His central venous pressure rose by 4 mmHg and his cardiac output rose by 0.2 L/min. Using the Frank-Starling curve you drew last week, predict what a second 500 mL would have done. Then say what happened to his oxygen saturation in those twenty-five minutes and name the force that changed to produce it.

3. Explain the baroreflex: name the sensor, where it sits, the afferent nerve, the integrator, and both efferent limbs. Then say what carvedilol does to that loop, and use it to explain why a heart rate of 118 in him is a larger signal than a heart rate of 118 in a man who takes nothing.

4. Interpret a mean arterial pressure of 73 at 1100. Say what it shows and what it cannot show about whether his kidneys and his gut are being perfused, and name the numbers on this page that argue they were not.

5. Name the one measurement that would tell you whether to add more noradrenaline or to add dobutamine, and say what a high and a low result would each mean.

FROM YOUR DESK

Your entry point's data and question are on [this week's entry page](#). Answer it here.

STUDENT: _____ ENTRY POINT: _____

11 The infection, and the response to it

November 4 to November 8 · Day 1 to day 5 · Blood, and How the Body Defends Itself

Two things are happening to him and they need separating. There is an organism, and there is what his body is doing about the organism. The antibiotic treats the first one. Almost everything on this page, including the thing that nearly killed him, is the second one. By Day 3 his white cell count has fallen below normal without anyone treating it, and that is worse news than the 19.4 he arrived with.

RESULTS RELEASED THIS WEEK

COMPLETE BLOOD COUNT, DAY 1

Collected November 4

Test	Result	Units	Reference range	Flag
White cells	19.4	$\times 10^9/L$	4.0 to 11.0	High
Neutrophils	88	%	40 to 70	High
Band forms	12	%	0 to 5	High
Lymphocytes	6	%	20 to 40	Low
Hemoglobin	11.8	g/dL	13.5 to 17.5	Low
Hematocrit	35	%	41 to 53	Low
Platelets	160	$\times 10^9/L$	150 to 400	

COMPLETE BLOOD COUNT, DAY 3

Collected November 6

Test	Result	Units	Reference range	Flag
White cells	3.1	$\times 10^9/L$	4.0 to 11.0	Low
Neutrophils	62	%	40 to 70	
Band forms	18	%	0 to 5	High
Lymphocytes	14	%	20 to 40	Low
Hemoglobin	9.6	g/dL	13.5 to 17.5	Low
Hematocrit	29	%	41 to 53	Low
Platelets	84	$\times 10^9/L$	150 to 400	Low

COAGULATION AND INFLAMMATION

Collected Day 1 and Day 3

Test	Result	Units	Reference range	Flag
INR, Day 1	1.2		0.8 to 1.2	
INR, Day 3	1.8		0.8 to 1.2	High
Fibrinogen, Day 3	180	mg/dL	200 to 400	Low
D-dimer, Day 3	6,400	ng/mL	under 500	High
C-reactive protein, Day 3	240	mg/L	under 5	High
Procalcitonin, Day 1	8.4	ng/mL	under 0.5	High
Procalcitonin, Day 4	2.1	ng/mL	under 0.5	High

YOUR ENTRIES

NEW ON THE CHART THIS WEEK

Copy the numbers into the flowsheets first, then list here what changed since the last column.

PROBLEM LIST, ADDITIONS AND CHANGES

Refer to problems by the numbers you gave them on the problem list.

DRAW THE CONTROL LOOP

Regulated variable, sensor, integrator, effector, and where this week it is failing or being pushed.

THE FIVE QUESTIONS

1. Name each cell line in a differential count and say what each one does. Then say which line is raised on Day 1, which lines have fallen by Day 3, and which single cell line is doing most of the damage to him.
2. His white cell count fell from 19.4 to 3.1 with nothing done to it. Give two mechanisms that could produce that fall, say what each one predicts about the bone marrow, and say which one his band count of 18 percent argues for.
3. Explain what a fever is in control system terms: name the regulated variable, the sensor, the integrator, the effectors, and say precisely what changed. Then say what a temperature of 35.8 °C (96.4 °F) on Day 3, in a man with live bacteria in his blood, means about that loop.
4. Interpret a platelet count of 84 with an INR of 1.8, a fibrinogen of 180 and a D-dimer of 6,400, in a man who is not bleeding. Say what that pattern shows and what it cannot show.
5. Name the one measurement that would tell you whether his fall in hemoglobin from 11.8 to 9.6 is dilution, bleeding, or something being made less of, and say what each result would look like.

FROM YOUR DESK

Your entry point's data and question are on this week's entry page. Answer it here.

STUDENT: _____ ENTRY POINT: _____

12 Nutrition, metabolism and the gut

November 8 to November 16 · Day 5 to day 13 · Digestion, and How You Use Food for Fuel

He is not eating and he is burning about 600 kcal a day more than a man his size at rest should. The fuel is coming from him. This entry is about where it is coming from, why feeding him is a physiological intervention rather than hotel service, and why the first few days of feeding a starved patient are more dangerous than the starving was.

RESULTS RELEASED THIS WEEK

REFEEDING ELECTROLYTES

Collected Day 4 to Day 6

Test	Result	Units	Reference range	Flag
Phosphate, Day 4	2.4	mg/dL	2.5 to 4.5	Low
Phosphate, Day 6	1.4	mg/dL	2.5 to 4.5	Critical low
Magnesium, Day 6	1.4	mg/dL	1.7 to 2.4	Low
Potassium, Day 6	3.2	mEq/L	3.5 to 5.0	Low
Glucose, Day 6	246	mg/dL	70 to 99	High

NUTRITION AND LIVER MARKERS

Collected Day 5

Test	Result	Units	Reference range	Flag
Albumin, Day 1	2.9	g/dL	3.5 to 5.0	Low
Albumin, Day 5	2.2	g/dL	3.5 to 5.0	Low
Prealbumin	9	mg/dL	18 to 38	Low
C-reactive protein	240	mg/L	under 5	High
Total bilirubin	2.4	mg/dL	0.2 to 1.2	High
ALT	96	U/L	10 to 40	High
Alkaline phosphatase	180	U/L	40 to 130	High
Hemoglobin A1c, on admission	7.9	%	under 5.7	High

YOUR ENTRIES

NEW ON THE CHART THIS WEEK

Copy the numbers into the flowsheets first, then list here what changed since the last column.

PROBLEM LIST, ADDITIONS AND CHANGES

Refer to problems by the numbers you gave them on the problem list.

DRAW THE CONTROL LOOP

Regulated variable, sensor, integrator, effector, and where this week it is failing or being pushed.

THE FIVE QUESTIONS

1. Trace one gram of the protein in his nasogastric feed from the tip of the tube to the inside of a muscle cell. Name the enzyme, the site and the transporter at every step, and say which of those steps a sedated ventilated patient does badly.
2. His measured energy expenditure is 2,240 kcal/day against a predicted 1,650. Say where the extra 590 is going and name the three hormones driving it. Then calculate his nitrogen balance from the Day 7 figures and use it to predict how much lean tissue he loses over ten days at that rate.
3. Explain why his glucose is running near 250 on an insulin infusion when he is being given almost no carbohydrate. Name the tissue making the glucose, the substrate it is made from, and the hormones telling it to. Then say why giving him more carbohydrate would not fix it.
4. Interpret an albumin of 2.2 g/dL. Say what it shows and what it cannot show about his nutrition, and name the two processes other than intake that moved it.
5. Name the one measurement that would tell you whether his gut is working, and say what you would do differently for each answer.

FROM YOUR DESK

Your entry point's data and question are on this week's entry page. Answer it here.

STUDENT: _____ ENTRY POINT: _____

13 Respiratory failure, ventilation and weaning

November 5 to November 10 · Day 2 to day 7 · Breathing, Gas Transport, and the Fast pH Lever

His lung fails in two stages and they are not the same problem. First it gets wet, because of what you worked on in Week 10. Then it gets stiff, because of what you worked on in Week 11. By Day 2 the fast lever that normally holds a pH steady within a minute is being asked to cover for a kidney that has stopped working, and the muscles doing the levering are the same ones that wasted in Week 12.

RESULTS RELEASED THIS WEEK

GAS EXCHANGE, DERIVED

Collected Day 1 and Day 2

Test	Result	Units	Reference range	Flag
PaO ₂ to inspired oxygen ratio, Day 1	141	mmHg	over 300	Low
PaO ₂ to inspired oxygen ratio, Day 2, 0400	73	mmHg	over 300	Low
PaO ₂ to inspired oxygen ratio, Day 2, 0800	111	mmHg	over 300	Low
PaO ₂ to inspired oxygen ratio, Day 4	167	mmHg	over 300	Low

RESPIRATORY MECHANICS, DAY 2 VENTILATED

Collected November 5, 0800

Test	Result	Units	Reference range	Flag
Tidal volume	440	mL	6 mL per kg predicted body weight	
Plateau pressure	28	cmH ₂ O	under 30	
PEEP	12	cmH ₂ O	set	
Driving pressure	16	cmH ₂ O	under 15	High
Static compliance	27.5	mL/cmH ₂ O	50 to 100	Low

DAY 7 BREATHING TRIAL

Collected November 10

Test	Result	Units	Reference range	Flag
Respiratory rate	34	breaths/min	12 to 20	High
Tidal volume	260	mL	about 440 set on the ventilator	Low
Rapid shallow breathing index	131	breaths/min/L	under 105 predicts success	High
Anatomic dead space, estimated	147	mL	about 2 mL per kg predicted body weight	

YOUR ENTRIES

NEW ON THE CHART THIS WEEK

Copy the numbers into the flowsheets first, then list here what changed since the last column.

PROBLEM LIST, ADDITIONS AND CHANGES

Refer to problems by the numbers you gave them on the problem list.

DRAW THE CONTROL LOOP

Regulated variable, sensor, integrator, effector, and where this week it is failing or being pushed.

THE FIVE QUESTIONS

1. Calculate his arterial oxygen content on Day 2 at 0400 and at his March baseline, using 1.34 mL of oxygen carried per gram of hemoglobin and 0.003 mL per mmHg dissolved. Show both. Say which of the two terms does almost all the work, and what that means about treating him with oxygen alone.
2. His alveolar to arterial oxygen difference is very large and it barely improves when the inspired oxygen is raised from 0.44 to 0.80. Name the two mechanisms that produce a low arterial oxygen with a large alveolar to arterial difference, say exactly how raising inspired oxygen separates them, and say which one he has.
3. Explain the fast pH lever: name the reaction, the enzyme, the organ, and say how a change in breathing changes blood pH within a minute. Then apply Winter's formula to his Day 2, 0400 gas and say whether his lever is working. Say what that answer told the team to do at 0510.
4. Interpret a plateau pressure of 28 with a PEEP of 12 and a tidal volume of 440 mL. Calculate his compliance, say what it shows, and say what it cannot show about whether the stiffness is in his lung or in his chest wall.
5. Name the one measurement that would tell you whether his failed breathing trial on Day 7 is a lung problem or a pump problem, and say what each result would mean.

FROM YOUR DESK

Your entry point's data and question are on this week's entry page. Answer it here.

STUDENT: _____ ENTRY POINT: _____

14 Acute kidney injury on chronic disease

November 4 to November 18 · Day 1 to day 15 · The Kidney and Body Fluid Balance

Everyone says his kidney failed because it was not getting enough blood. Look at his central venous pressure of 14 and ask what a filter does when the pressure downstream of it rises. This is the week where the number people forget turns out to be the one doing the damage, and where a creatinine that improves at the end of the file improves partly for a reason that has nothing to do with his kidney.

RESULTS RELEASED THIS WEEK

URINE INDICES, DAY 1

Collected November 4

Test	Result	Units	Reference range	Flag
Urine sodium	12	mEq/L	read against the clinical state	
Urine creatinine	59	mg/dL		
Fractional excretion of sodium	0.4	%	under 1 suggests an intact tubule	Low
Urine osmolality	520	mOsm/kg	a concentrating tubule can exceed 600	
Urine microscopy	no casts			

URINE INDICES, DAY 3

Collected November 6

Test	Result	Units	Reference range	Flag
Urine sodium	48	mEq/L	read against the clinical state	
Urine creatinine	52	mg/dL		
Fractional excretion of sodium	2.6	%	over 2 suggests tubular injury	High
Urine osmolality	310	mOsm/kg	plasma osmolality is about 300	
Urine microscopy	muddy brown granular casts		none	High

RENAL CHEMISTRY OVER THE ADMISSION

Collected Day 1 to Day 15

Test	Result	Units	Reference range	Flag
Creatinine, Day 1	2.6	mg/dL	baseline 1.5	High
Creatinine, Day 3	3.8	mg/dL	baseline 1.5	High
Creatinine, Day 15	2.4	mg/dL	baseline 1.5	High
Estimated glomerular filtration rate, Day 15	29	mL/min/1.73 m ²	baseline 50	Low
Potassium, Day 3	5.6	mEq/L	3.5 to 5.0	High
Potassium, Day 8	6.2	mEq/L	3.5 to 5.0	Critical high
Bicarbonate, Day 8	13	mEq/L	22 to 29	Critical low
Albumin, Day 5	2.2	g/dL	3.5 to 5.0	Low

YOUR ENTRIES

NEW ON THE CHART THIS WEEK

Copy the numbers into the flowsheets first, then list here what changed since the last column.

PROBLEM LIST, ADDITIONS AND CHANGES

Refer to problems by the numbers you gave them on the problem list.

DRAW THE CONTROL LOOP

Regulated variable, sensor, integrator, effector, and where this week it is failing or being pushed.

THE FIVE QUESTIONS

1. Name the three pressures that set glomerular filtration and say which way each one moved in him on Day 1. Then name the one people forget, and use his central venous pressure of 14 to explain how a filter can fail because of what is happening downstream of it.
2. His fractional excretion of sodium was 0.4 percent on Day 1 and 2.6 percent on Day 3, and his urine osmolality fell from 520 to 310 mOsm/kg. Say what each pair of numbers reports about the tubule, and name what changed between the two days.
3. Explain how the kidney defends blood pressure and volume: name the sensor, the enzyme it releases, the two hormones that follow, and the tubule segment each one acts on. Then say what holding his lisinopril and his spironolactone on Day 1 did to that system, and why it was held anyway.
4. Interpret a creatinine of 2.6 on Day 1 in a man whose baseline is 1.5. Say what it shows and what it cannot show about how much filtration he had actually lost by that morning, and explain why.
5. Name the one measurement that would tell you whether more diuretic or less fluid is the right move on Day 6, and say what a high and a low result would each mean.

FROM YOUR DESK

Your entry point's data and question are on this week's entry page. Answer it here.

STUDENT: _____ ENTRY POINT: _____

15 Integration

Both files, read again · November 4 again, and September 22 again · The Slow pH Lever, and Putting It Together

This is the last entry and it is the first one again, twice. You read Camila's arrival gas in Week 3 and were told to leave it alone. You read Dale's in Week 9 and were told the same. Read both of them now. Then read the gas each of them had once treatment had been running, which is the one that catches people out, because in both patients the treatment created a second acidosis while it was fixing the first.

RESULTS RELEASED THIS WEEK

ANION GAPS, FOR YOU TO CHECK

Collected Both patients

Test	Result	Units	Reference range	Flag
Camila, arrival, albumin 4.2	you calculate	mEq/L	normal about 12	
Camila, 1800, albumin 3.9	you calculate	mEq/L	normal about 12	
Dale, arrival, albumin 2.9	you calculate	mEq/L	normal about 12	
Dale, Day 7, albumin 2.1	you calculate	mEq/L	normal about 12	
Dale, Day 15, albumin 2.6	you calculate	mEq/L	normal about 12	

YOUR ENTRIES

NEW ON THE CHART THIS WEEK
Copy the numbers into the flowsheets first, then list here what changed since the last column.

PROBLEM LIST, ADDITIONS AND CHANGES
Refer to problems by the numbers you gave them on the problem list.

DRAW THE CONTROL LOOP

Regulated variable, sensor, integrator, effector, and where this week it is failing or being pushed.

THE FIVE QUESTIONS

1. Name the primary disturbance in each of the six blood gases on this page. For each one say which lever, the fast one or the slow one, is doing the compensating, and whether it is doing enough. Show the Winter's check for every gas where a metabolic acidosis is primary.

2. Calculate the anion gap for Dale on November 4 and on November 10, and correct both for his albumin. Say what the two corrected numbers, read with his chloride, reveal about what happened between those dates, and name the fluid responsible.
3. Explain the slow lever: name where filtered bicarbonate is reclaimed, where new bicarbonate is made, what the acid is excreted with, and how long the whole response takes. Then say why Dale's slow lever could not help him on November 4 and could not help him on November 10, for two completely different reasons.
4. Interpret Camila's 1800 gas and Dale's November 10 gas side by side. Both patients are improving. Both have a new acidosis that was made by the treatment. Say what each gas shows, what neither can show, and what you would change about the fluid in each case.
5. Both files open with a person who was well and one sentence of history. Name the one measurement you would have added to Camila's August physical and the one you would have added to Dale's March clinic visit, and for each say what question it would have let you answer later.

FROM YOUR DESK

Your entry point's data and question are on this week's entry page. Answer it here.

BOTH PATIENTS · BIO 005 PATIENT CHART

STUDENT: _____ ENTRY POINT: _____

The end of term analysis

One analysis, both patients, due in Week 15. It is written from this book. Nothing else is needed and nothing else is allowed as the source of a number.

THE ONE IDEA THAT READS BOTH FILES

Draft it here. It has to be a claim about physiology, not about either patient.

THREE NUMBERS FROM CAMILA THAT SUPPORT IT

Week, value, and what it is evidence of.

THREE NUMBERS FROM DALE THAT SUPPORT IT

Week, value, and what it is evidence of.

WHERE YOU CHANGED YOUR MIND

Which week you were wrong, which week you noticed, and what changed it. This is graded and it is worth real marks.

The full assignment, the structure and the rubric are on the analysis page.